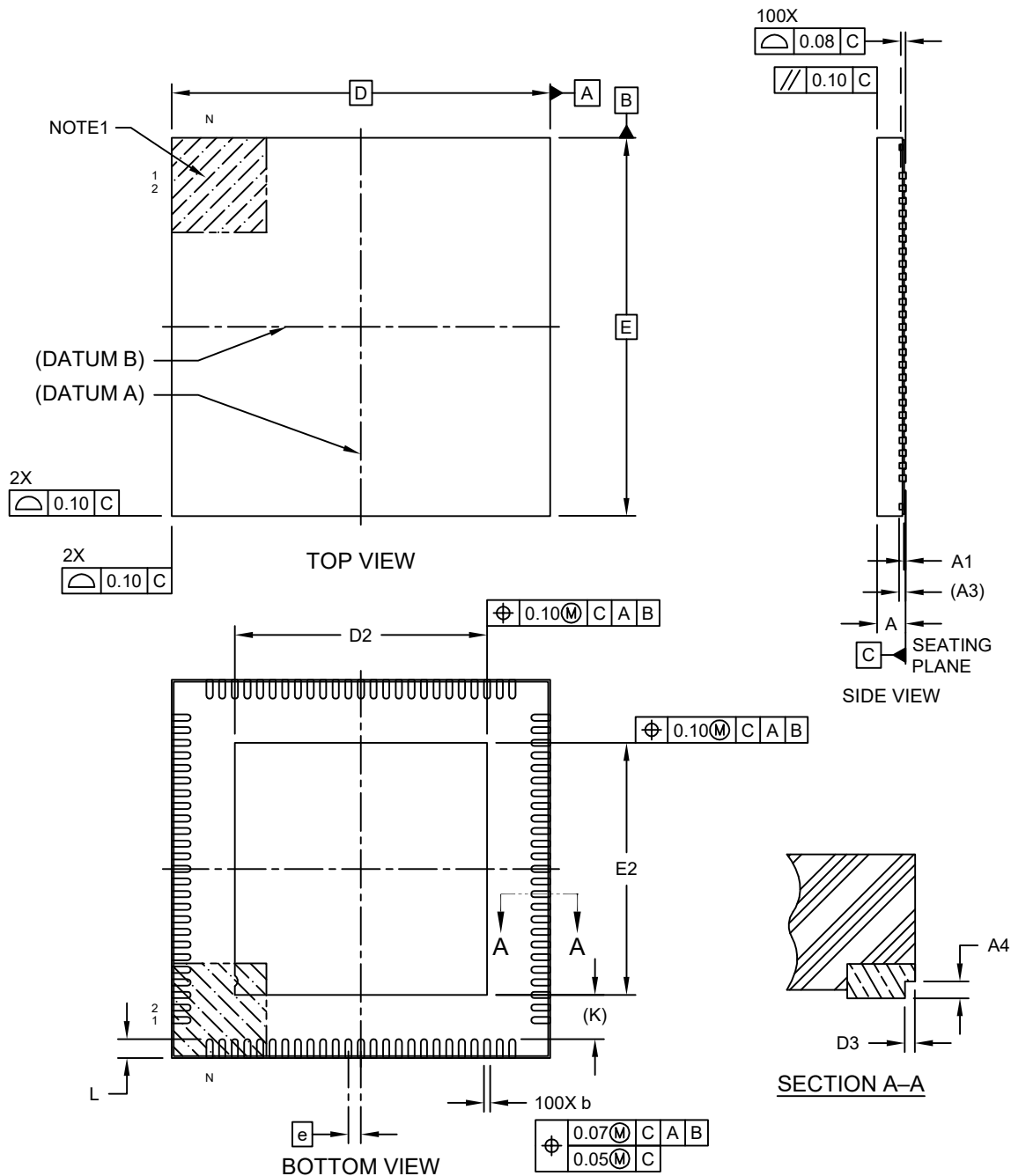


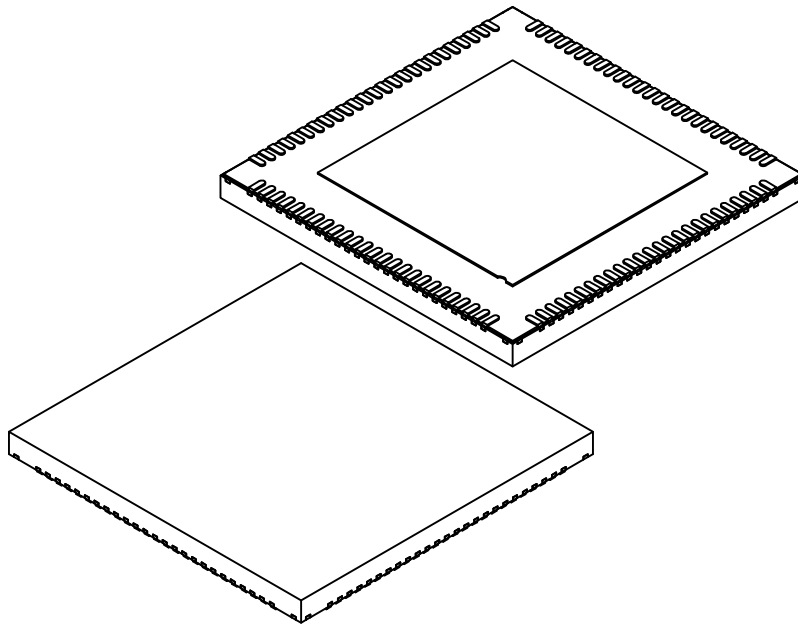
**100-Lead Very Thin Quad Flat, No Lead Package (ZVX) - 12x12x0.9 mm Body [VQFN]  
With 8.0 mm Exposed Pad and Stepped Wettable Flanks**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**100-Lead Very Thin Quad Flat, No Lead Package (ZVX) - 12x12x0.9 mm Body [VQFN]  
With 8.0 mm Exposed Pad and Stepped Wettable Flanks**

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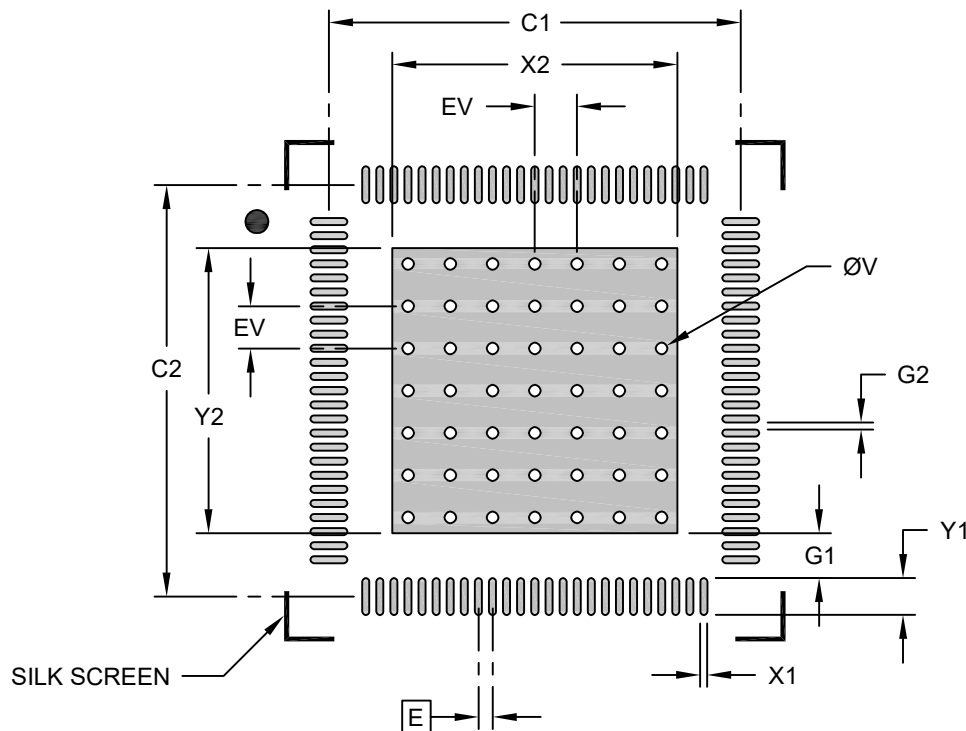
|                                |    | Units | MILLIMETERS |      |       |
|--------------------------------|----|-------|-------------|------|-------|
| Dimension Limits               |    |       | MIN         | NOM  | MAX   |
| Number of Terminals            | N  |       | 100         |      |       |
| Pitch                          | e  |       | 0.40 BSC    |      |       |
| Overall Height                 | A  |       | –           | –    | 0.90  |
| Standoff                       | A1 |       | 0.00        | 0.02 | 0.05  |
| Terminal Thickness             | A3 |       | 0.203 REF   |      |       |
| Overall Length                 | D  |       | 12.00 BSC   |      |       |
| Exposed Pad Length             | D2 |       | 7.90        | 8.00 | 8.10  |
| Overall Width                  | E  |       | 12.00 BSC   |      |       |
| Exposed Pad Width              | E2 |       | 7.90        | 8.00 | 8.10  |
| Terminal Width                 | b  |       | 0.15        | 0.20 | 0.25  |
| Terminal Length                | L  |       | 0.50        | 0.60 | 0.70  |
| Terminal-to-Exposed-Pad        | K  |       | 1.40 REF    |      |       |
| Wettable Flank Step Cut Length | D3 |       | –           | –    | 0.085 |
| Wettable Flank Step Cut Height | A4 |       | 0.10        | –    | 0.19  |

**Notes:**

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
  - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
  - REF: Reference Dimension, usually without tolerance, for information purposes only.

# **100-Lead Very Thin Quad Flat, No Lead Package (ZVX) - 12x12x0.9 mm Body [VQFN] With 8.0 mm Exposed Pad and Stepped Wettable Flanks**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## **RECOMMENDED LAND PATTERN**

| Units                            |    | MILLIMETERS |       |      |
|----------------------------------|----|-------------|-------|------|
| Dimension Limits                 |    | MIN         | NOM   | MAX  |
| Contact Pitch                    | E  | 0.40 BSC    |       |      |
| Center Pad Width                 | X2 |             |       | 8.10 |
| Center Pad Length                | Y2 |             |       | 8.10 |
| Contact Pad Spacing              | C1 |             | 11.70 |      |
| Contact Pad Spacing              | C2 |             | 11.70 |      |
| Contact Pad Width (Xnn)          | X1 |             |       | 0.20 |
| Contact Pad Length (Xnn)         | Y1 |             |       | 1.05 |
| Contact Pad to Center Pad (Xnn)  | G1 | 1.28        |       |      |
| Contact Pad to Contact Pad (Xnn) | G2 | 0.20        |       |      |
| Thermal Via Diameter             | V  |             | 0.33  |      |
| Thermal Via Pitch                | EV |             | 1.20  |      |

### **Notes:**

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-2529 Rev A